BITTENSOR SUBNET

ADVERSARIAL INFRASTRUCTURE

ARES

Decentralized Adversarial Robustness Infrastructure for Autonomous AI Agents

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Vision

Robustness Must Become a Market

AI agents are gaining economic agency.

They are entering:

- DeFi systems
- DAO governance
- Enterprise automation
- Financial markets

But robustness evaluation remains:

- Centralized
- Static
- Episodic
- Non-incentivized

> **Adversarial stress testing must become continuous, competitive, and economically incentivized.**

ARES operationalizes this shift.

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The Problem

The Missing Infrastructure Layer

Autonomous AI systems can:

- Lose capital
- Capture governance
- Manipulate markets
- Drift from objectives

Current robustness evaluation:

- Conducted internally
- Performed sporadically
- Not economically aligned
- Not transparent

There is no decentralized robustness oracle.

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The Solution

ARES: A Decentralized Red-Team Marketplace

ARES creates a co-evolutionary market between:

Miners

Autonomous agents attempting to complete tasks robustly.

Validators

Adversarial environment generators attempting to break them.

Emissions flow to:

Agents that remain robust under adversarial pressure.

> **ARES transforms robustness into a measurable economic signal.**

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Why Now

AI Agents Are Entering Financial Systems

Catalysts:

- Rapid rise of agentic AI frameworks
- Autonomous trading systems
- DAO automation tools
- On-chain AI integrations

Risk surface expanding faster than evaluation infrastructure.

> The economic stakes of AI robustness are increasing — but evaluation mechanisms have not evolved.

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Market Opportunity

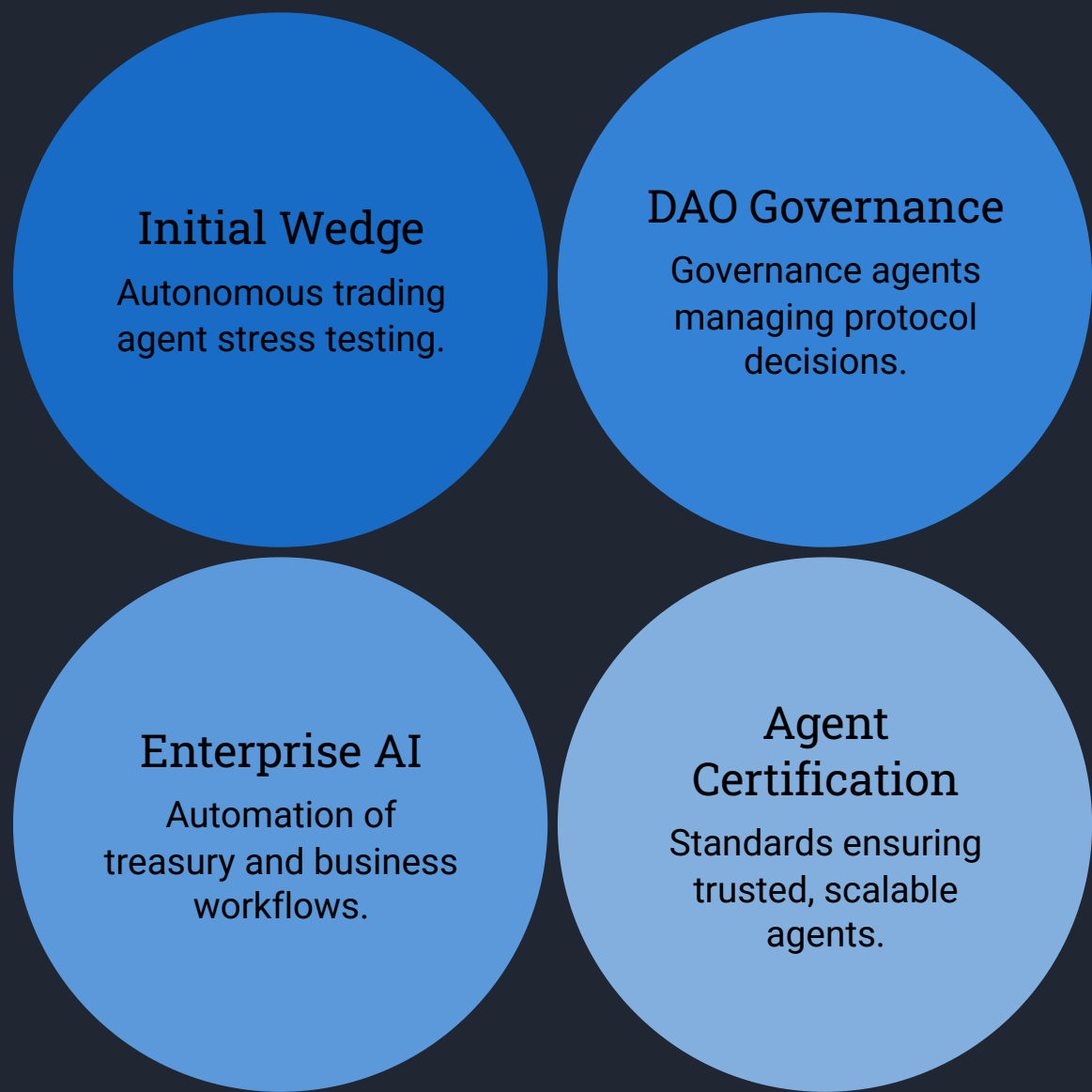
Robustness as Infrastructure

Initial wedge:

Autonomous trading agent stress testing

Expansion vectors:

- DAO governance agents
- Enterprise AI automation
- AI-powered treasury systems
- Agent certification layers



Long-term opportunity: ARES becomes the default decentralized robustness benchmark.

Position ARES as: **Infrastructure, not a tool.**

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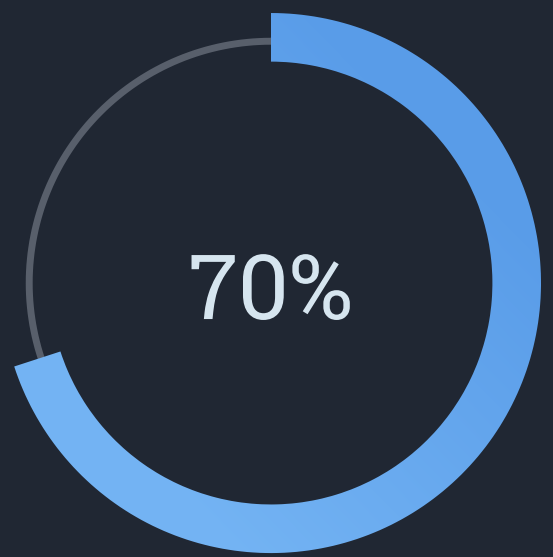
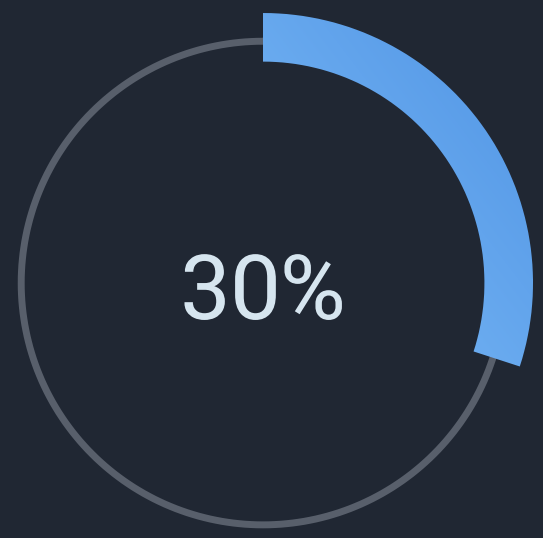
Mechanism Design Advantage

Incentive-Aligned by Construction

Emission split:



Miners



Validators

Miners rewarded for:

- Accuracy
- Robustness
- Goal fidelity
- Stability
- Efficiency

Validators rewarded for:

- Exploit impact
- Novelty
- Consensus agreement
- Low false positives

Median aggregation + randomized assignment reduces collusion risk.

ARES embeds adversarial pressure directly into reward logic.

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Why Bittensor

Native to Bittensor Architecture

ARES requires:

- Dual-role competition
- Continuous ranking
- Emission-weighted incentives
- Adaptive weight updates

Bittensor uniquely provides:

- Miner-validator market structure
- Transparent emissions
- Open participation
- Incentive-driven evolution

No conventional blockchain supports co-evolutionary intelligence markets.

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Defensibility

Structural Moats

1 **Co-evolutionary data flywheel**

More attacks → more robustness → higher signal quality.

2 **Historical credibility weighting**

Long-term validator reputation compounds.

3 **Execution trace dataset**

High-value behavioral robustness data accrues over time.

4 **Network effects**

More miners and validators increase evaluation quality.

ARES strengthens as it grows.

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ARES as a Robustness Oracle

Bootstrapping Strategy

Initial target: **Crypto-native AI builders.**
Anchor use case: **Autonomous trading stress testing.**

Goal: **Seed high-quality adversarial pressure early.**

Miners
Early emission multipliers
Governance influence

Validators
Exploit discovery bonuses
Leaderboards

Users
Free API credits
Public robustness dashboard

Future evolution:

1. Decentralized robustness certification
2. Cross-subnet robustness scoring
3. Insurance-grade evaluation signal
4. Agent risk assessment primitive

> **Only agents that withstand adversarial pressure should receive economic reward.**
>
> **ARES makes robustness measurable, competitive, and decentralized.**

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